TEST REPORT NO.:ETX202303-07-078-01

Standard	IEC 60601-1-2:2014+A1:2020
Client **Address**	Andon Health Co., Ltd. No.3 JinPing Street, Ya An Road, nankai District, Tianjin, China
Trade Mark	N/A
Test Item	Infrared No-Touch Forehead Thermometer
Model/type reference	PT9L
Serial Number	KE04(LOT)
Test laboratory **Address**	Tianxu(Tianjin)Testing Technology Service Co.,LTD 101, First floor, Building L,No.3 JinPing Street,Ya An Road, Nankai District, Tianjin,China
Testing period	2024.07.05-2024.07.29
Test Result	The test item passed the test specification(s).
Prepared by	2024-08-15 (Date) Sun Liang (Name) signature
Reviewed by	2024-08-15 (Date) Zhang Le (Name) signature
Other aspects	N/A

Abbreviation

P(ass) :passed F(ail) : failed

N/A: not applicable N/T: not tested

Note

This test report relates to the a. m. test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any safety mark on this or similar products.

1 Test Summary

Electromagnetic Interference(EMI)			
Test	**Requirement**	**Test Method**	**Result**
Conducted Emission (150 kHz to 30 MHz)	IEC 60601-1-2:2014+A1:2020	CISPR11group 1, class B	N/A
Radiated Emission (30MHz to 1000MHz)	IEC 60601-1-2:2014+A1:2020	CISPR11 group 1, class B	PASS
Electromagnetic Susceptibility(EMS)			
Test	**Requirement**	**Test Method**	**Result**
Electrostatic Discharge (ESD)	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-2	PASS
Radiated RF Electromagnetic Fields	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-3	PASS
Proximity field from RF wireless communication equipment.	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-3	PASS
Electrical Fast Transients and Bursts (EFT/B)	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-4	N/A
Surges	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-5	N/A
Conducted Disturbance, Induced by RF Fields (150 kHz to 80 MHz)	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-6	N/A
Voltage Dips, Short Interruptions and Voltage Variations	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-11	N/A
Power Frequency Magnetic Field	IEC 60601-1-2:2014+A1:2020	IEC 61000-4-8	Exemption, see annex

声明

致天旭（天津）检测技术服务有限公司:

我司生产的 PT9L 红外体温计(含主机、随机附件)符合 IEC 60601-1-2:2014/AMD1:2020 中 8.11a 条款中的豁免条件，外壳内不包含磁敏感元器件及电路，故特此声明。

天津九安医疗电子股份有限公司

2024 年 07 月 29 日

2 General Product Information

2.1 General Description of E.U.T.

Product Name:	Infrared No-Touch Forehead Thermometer
Sample type:	☒ Client provided ☐ Random sampling
Model No.(EUT):	PT9L
Dimensions (W x H x D)	5.15 in x 1.37 in x 1.49 in
Applicant:	Andon Health Co., Ltd.
Address of Applicant:	No. 3 JinPing Street, YaAn Road, nankai District, Tianjin 300190, China
Manufacturer:	Andon Health Co., Ltd.
Address of Manufacturer:	No. 3 JinPing Street, YaAn Road, nankai District, Tianjin 300190, China
Rated Input:	DC 3V, 2*AAA battery
AC-DC adapter	N/A
Attachment	N/A
Software version	PT-HT2762-V00029-1.0.0
Firmware version	PT9L-HFMP01 V1.0
Intended healthcare environment	Professional healthcare facility environment and HOME HEALTHCARE ENVIRONMENT as described in IEC 60601-1-2 (ed.4.1)

2.2 Product Intended Use and intended environments

INTENDED USE:

This product is a high-tech infra-red (IR) thermometer designed to take human body temperature by measuring the energy of IR emitted from the forehead. The product will help you to assess you and your family members' health conditions easily and quickly.

Intended environments:

Professional healthcare facility environment and HOME HEALTHCARE ENVIRONMENT as described in IEC 60601-1-2 (ed.4.1)

2.3 Submitted Documents

Circuit diagram, PCB layout, components list, user's manual and label.

2.4 Deviation from Standards

None.

2.5 Abnormalities from Standard Conditions

None.

2.6 Modification/Retest record

None.

2.7 Countermeasures to achieve EMC Compliance

None.

3 Test Set-up and Operation Mode

3.1 E.U.T Mode

Mode item	Mode Name	Mode description
Mode 1	Battery power Standby mode	Install battery to the EUT and keep the EUT in standby state
Mode 2	Battery power Test mode	PT9L is powered by the internal battery. Hold down the power button to install the battery and set PT9L in UE1 mode. During the radiation disturbance test, PT9L detects the air and carries out continuous measurement. During the immunity test, the temperature sensor detection is point to the blackbody,and the blackbody is set to 37℃.The temperature is continuously measured and compared with the blackbody temperature. The measurement accuracy is ±0.2℃.

Emission: The equipment under test (EUT) was configured to measure its highest possible emission level. The test conditions were adapted accordingly in reference to the instructions for use.

Immunity: The equipment under test (EUT) was configured to have its highest possible susceptibility against the tested phenomena. The test conditions were adapted accordingly in reference to the instructions for use.

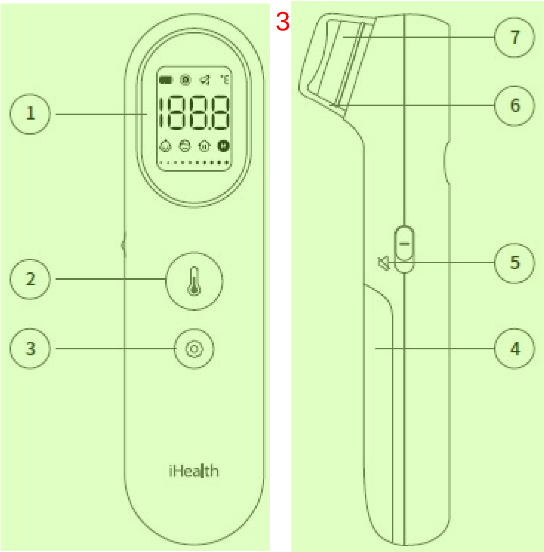
Refer to the related paragraph of this report for more information.

3.2 Description of Support Units and Auxiliary Equipment

Equipment	Manufacturer	Model NO.
BDB Series Precision Temperature Controllers & Blackbody Radiation Source	Beijing Boda Changzheng Technology Development Co., LTD	BDB15

3.3 Test setup electrical and physical diagrams （Block diagram of the ME equipment and ME system and all peripherals and auxiliary equipment used）

Connection diagram



① LCD display

② Measurement button

③ Settings button

④ Battery cover

⑤ Mute switch

⑥ Protective cap

⑦ Probe

3.4 Measurement Uncertainty

According to CISPR 16-4-2.

Test Item	Frequency Range	Measurement Uncertainty	U_{cispr}
Conducted Emission at mains port using AMN	150kHz-30MHz	3.2dB	3.4dB
Radiated Emission	30MHz-1000MHz	3.9dB	6.3dB
Radiated Emission	1GHz-6GHz	5.1dB	5.2dB
Remark: AMN – Artificial Mains Network			

Note: The measurement uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

4 Equipment List

Radiated Emission

Item	Test Equipment	Manufacturer	Model No.	Serial No.	Cal. Date	Cal.Due date
1	Test receiver	Agilent	N9038A-508	MY54130074	2024-12-30	2025-12-29
2	310 Low Noise Amplifier	SONOMA	310N	344218	2024-01-08	2026-01-07
3	BiConiLog Antenna	ETS-Lindgren	3142E	166231	2024-05-11	2027-05-10

Electrostatic Discharge

Item	Test Equipment	Manufacturer	Model No.	Serial No.	Cal. Date	Cal.Due date
1	ESD simulator	EMTEST	Dito	P1420134242	2024-04-28	2025-04-27

Radiated RF Electromagnetic Fields

Item	Test Equipment	Manufacturer	Model No.	Serial No.	Cal. Date	Cal.Due date
1	Signal generator	Agilent	N5171B-506	MY53050921	2024-12-30	2025-12-29
2	RF power amplifier	Milmega	80RF1000-250	1066643	2024-01-08	2027-01-07
3	Directional coupler	Werlatone(HDC)	C5982-12	106464	NA	NA
4	RF power amplifier	Milmega	AS0827-55	1066644	2023-12-28	2026-12-27
5	Directional coupler	Werlatone(HDC)	C6148-12	106400	NA	NA
6	Power meter	Agilent	N1914A	MY54466075	2024-12-30	2025-12-29
7	Power sensor(2 set)	Agilent	E9304A	MY54370020/MY54370019	2024-12-30	2025-12-29
8	Isotropic field probe	ETS-Lindgren	HI6105-USB	168988	2024-05-24	2025-05-23
9	Log periodic antenna	Schwarzbeck	STLP9128D	9128D088	NA	NA
10	Log periodic antenna	Schwarzbeck	STLP9149	9149-465	NA	NA
11	RF power amplifier	BONN	BLMA 2060-100S	1610981	2023-01-14	2026-01-13

Environment

Item	Test Equipment	Manufacturer	Model No.	Serial No.	Cal. Date	Cal.Due date
1	semi-anechoic chamber	TDK	SAC-3	N/A	2024-01-05	2029-01-04

5 Electromagnetic Interference Test Results

5.1 Conducted Emissions on Mains Terminals, 150 kHz to 30 MHz

Test Requirement: IEC 60601-1-2:2014+A1:2020
Test Method: N/A
Test Date: N/A
Temperature and Humidity: N/A
Power Supply: N/A
Frequency Range: N/A
Class / Severity: N/A
Detector: N/A

Limit:

Frequency range(MHz)	**At mains Terminals** (dB μV)	
	Quasi-peak	Average
0.15 to 0.50	66-56	56-46
0.50 to 5	56	46
5 to 30	60	50
Note1: The lower limit is applicable at the transition frequency.		

Note: The EUT is powered by batteries , and does not apply to this clause.

5.2 Radiated Emissions, 30MHz to 1GHz

Test Requirement: IEC 60601-1-2:2014+A1:2020

Test Method: CISPR11

Test Date: 2024-07-05

Temperature and Humidity: 28℃ 52%

Power Supply: DC 3V

Frequency Range: 30 MHz to 1 GHz

Test site: 3m Semi-anechoic chamber

Class: Group 1, Class B

Detector: Peak for pre-scan (120 kHz resolution bandwidth)

Limit:

Frequency range(MHz)	**Quasi-peak limits dB (μV/m)**
30 to 230	40
230 to 1000	47
At transitional frequencies the lower limit applies.	

5.2.1 E.U.T. Operation

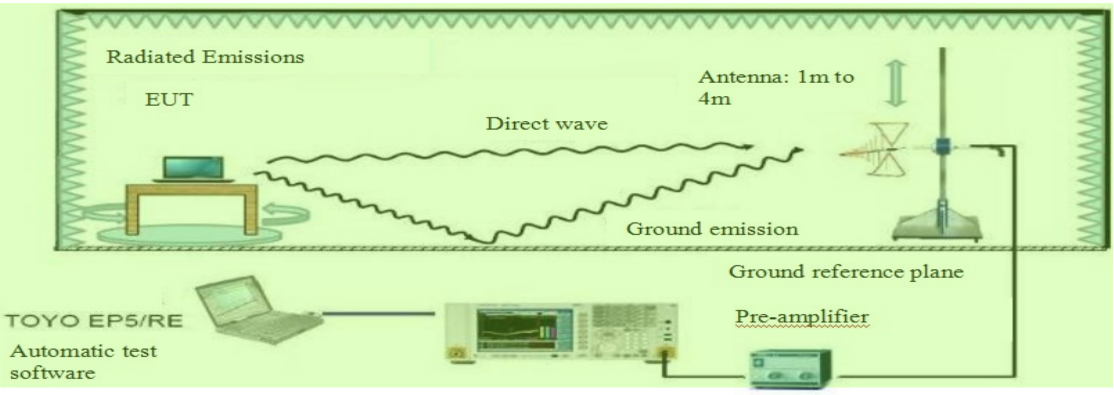
Test mode: Mode1,Mode2

Pre-scan was performed with peak detector on the EUT. Quasi-peak measurements were performed at the frequencies at which maximum peak emission level were detected. During the tests, the EUT were tested at different working modes to find the maxium emission results.

Please see the attached Quasi-peak test results.

For radiated emission: Level = Read Level + Antenna Factor + Cable Loss – Preamp Factor.

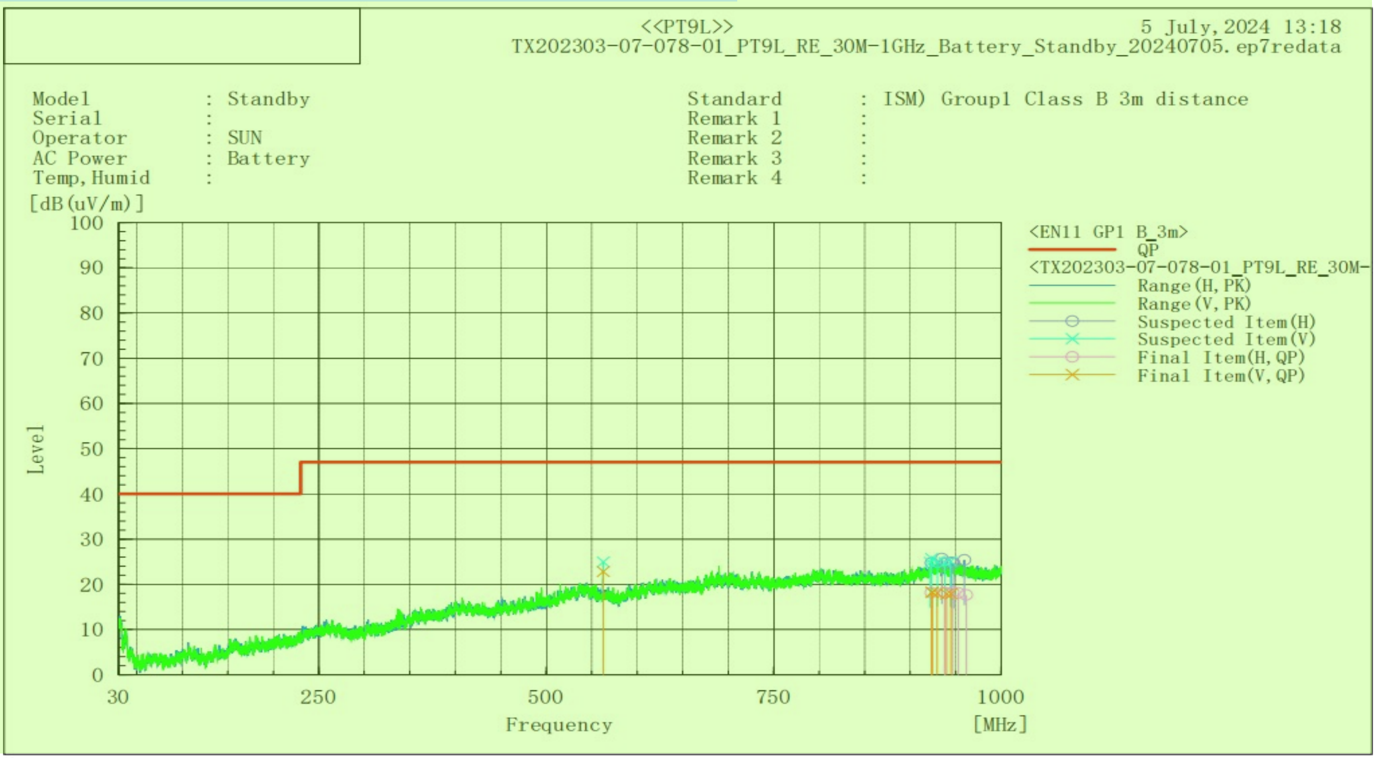
5.2.2 Test Setup and Procedure



1. The radiated emission test was conducted in a semi-anechoic chamber.
2. The tabletop EUT was placed upon a non-metallic table 0.8m above the ground reference plane. And for floor-standing arrangement, the EUT was placed on the horizontal ground reference plane, but separated from metallic contact with the ground reference plane by 0.1m of insulation.
3. Before final measurements of radiated emissions, a pre-scan was performed in the spectrum mode with the peak detector to find out the maximum emission spectrum signature data plots of the EUT.
4. The frequencies of maximum emission were determined in the final radiated emissions measurement, the physical arrangement of the test system and associated cabling was varied in order to determine the effect on the EUT's emissions in amplitude, direction and frequency. At each frequency, the EUT was rotated 360°, and the Antenna was raised and lowered from 1 to 4 meters in order to determine the maximum disturbance. Measurements were performed for both horizontal and vertical Antenna polarization.

5.2.3 Measurement Data

Figure 1: Measurement results of radiated emission, Mode 1



Final Result

--- Horizontal Polarization (QP)---

No.	Frequency [MHz]	Reading [dB(uV)]	c.f [dB(1/m)]	Result [dB(uV/m)]	Limit [dB(uV/m)]	Margin [dB]	Height [cm]	Angle [deg]	Remark
1	923.291	21.90	-3.74	18.16	47.00	28.84	180.50	222.80	
2	937.960	21.54	-3.37	18.17	47.00	28.83	308.20	84.60	
3	937.601	21.44	-3.36	18.08	47.00	28.92	214.50	44.10	
4	946.606	21.87	-3.60	18.27	47.00	28.73	387.50	359.30	
5	952.726	21.82	-3.74	18.08	47.00	28.92	364.80	71.30	
6	961.464	21.51	-3.87	17.64	47.00	29.36	399.60	86.30	

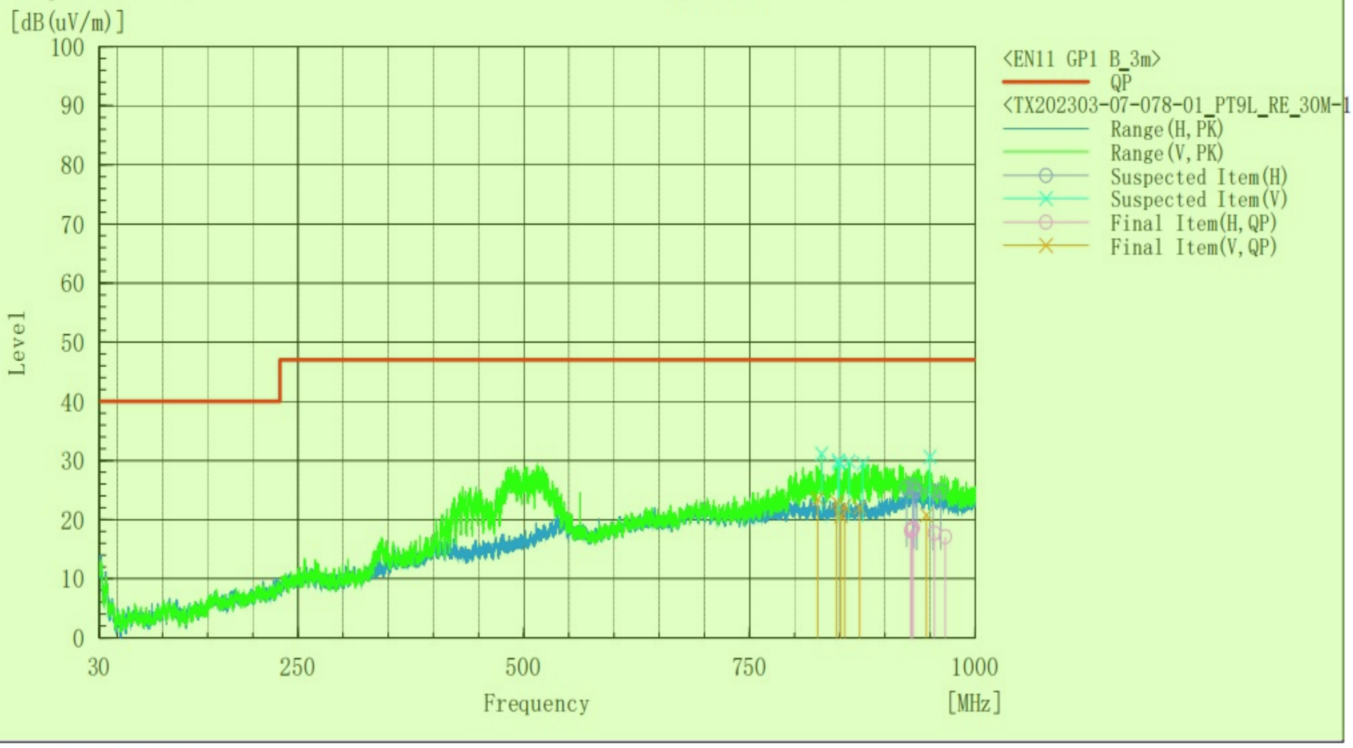
--- Vertical Polarization (QP)---

No.	Frequency [MHz]	Reading [dB(uV)]	c.f [dB(1/m)]	Result [dB(uV/m)]	Limit [dB(uV/m)]	Margin [dB]	Height [cm]	Angle [deg]	Remark
1	562.548	33.14	-10.29	22.85	47.00	24.15	180.80	222.60	
2	923.755	21.99	-3.72	18.27	47.00	28.73	399.70	341.20	
3	924.418	21.92	-3.69	18.23	47.00	28.77	396.40	330.80	
4	929.585	21.64	-3.53	18.11	47.00	28.89	139.90	86.10	
5	940.070	21.27	-3.41	17.86	47.00	29.14	392.10	80.30	
6	945.031	21.50	-3.55	17.95	47.00	29.05	267.60	33.00	

Figure 2: Measurement results of radiated emission, Mode 2

<<PT9L>> 5 July,2024 15:12
TX202303-07-078-01_PT9L_RE_30M-1GHz_Battery_Test_20240705.ep7redata

Model	: Test	Standard	: ISM) Group1 Class B 3m distance
Serial	:	Remark 1	:
Operator	: SUN	Remark 2	:
AC Power	: Battery	Remark 3	:
Temp,Humid	:	Remark 4	:



Final Result

--- Horizontal Polarization (QP)---

No.	Frequency [MHz]	Reading [dB(uV)]	c.f [dB(1/m)]	Result [dB(uV/m)]	Limit [dB(uV/m)]	Margin [dB]	Height [cm]	Angle [deg]	Remark
1	928.671	21.67	-3.56	18.11	47.00	28.89	212.80	0.90	
2	929.430	21.82	-3.54	18.28	47.00	28.72	135.10	264.60	
3	929.968	21.93	-3.52	18.41	47.00	28.59	238.00	310.50	
4	931.203	22.00	-3.47	18.53	47.00	28.47	107.50	79.50	
5	954.813	21.45	-3.75	17.70	47.00	29.30	396.70	58.60	
6	966.678	21.22	-4.14	17.08	47.00	29.92	160.00	217.40	

--- Vertical Polarization (QP)---

No.	Frequency [MHz]	Reading [dB(uV)]	c.f [dB(1/m)]	Result [dB(uV/m)]	Limit [dB(uV/m)]	Margin [dB]	Height [cm]	Angle [deg]	Remark
1	825.502	29.77	-6.25	23.52	47.00	23.48	116.80	258.20	
2	851.196	27.00	-6.39	20.61	47.00	26.39	116.50	33.90	
3	846.354	29.27	-6.34	22.93	47.00	24.07	111.90	110.80	
4	855.596	28.68	-6.36	22.32	47.00	24.68	112.90	116.10	
5	872.003	28.21	-6.16	22.05	47.00	24.95	112.50	103.10	
6	945.991	24.37	-3.58	20.79	47.00	26.21	107.70	184.70	

6 Electromagnetic Susceptibility Test Results

6.1 Performance Criteria Description in IEC 60601-1-2

The following are examples that can be used to develop pass/fail criteria. For ME EQUIPMENT and ME SYSTEMS with multiple functions, the pass/fail criteria should be applied to each function, parameter and channel.

Examples of test failures:

– malfunction;

– non-operation when operation is required;

– unwanted operation when no operation is required;

– deviation from normal operation that poses an unacceptable RISK to the PATIENT or OPERATOR;

– component failures;

– change in programmable parameters;

– reset to factory defaults (MANUFACTURER's presets);

– change of operating mode;

– a FALSE POSITIVE ALARM CONDITION;

– a FALSE NEGATIVE ALARM CONDITION (failure to alarm);

– cessation or interruption of any intended operation, even if accompanied by an ALARM signal;

– initiation of any unintended operation, including unintended or uncontrolled motion, even if accompanied by an ALARM signal;

– error of a displayed numerical value sufficiently large to affect diagnosis or treatment;

– noise on a waveform in which the noise would interfere with diagnosis, treatment or monitoring;

– artefact or distortion in an image in which the artefact would interfere with diagnosis, treatment or monitoring;

– failure of automatic diagnosis or treatment ME EQUIPMENT or ME SYSTEM to diagnose or treat, even if accompanied by an ALARM signal.

Example of performance during and after the applied testing stimulus required to pass the test:

– for a mammography system, the compression full release and associated command remains fully operational;

– for ULTRASOUND DIAGNOSTIC EQUIPMENT, the probe heating, dissipative power and temperature shall remain within specifications;

– safety-related functions perform as intended;

– false operation of alarms, "fail safe" modes and similar functions do not occur.

NOTE This might require performing the test twice – once to ensure the functions occur as expected and again to ensure they do not occur falsely.

Examples of acceptable degradation:

– an imaging system displays an image that could be altered, but in a way that would not affect the diagnosis or treatment;

– a heart rate monitor displays a heart rate that could be in error, but by an amount that is not clinically significant;

– a PATIENT monitor exhibits a small amount of noise or a transient on a waveform and the noise or transient would not affect diagnosis, treatment or monitoring.

Examples of ME EQUIPMENT and ME SYSTEMS with multiple functions:

– multi-parameter monitors;

– anaesthesia system with monitors;

– ventilators with monitors;

– multiple instances of the same function (e.g. multiple invasive blood pressure sensors).

Failure of therapy equipment to terminate a treatment at the intended time can be considered cessation or interruption of an intended operation related to ESSENTIAL PERFORMANCE. If the effect of the test signal on an ME EQUIPMENT or ME SYSTEM is so brief as to be transparent to the PATIENT or OPERATOR and does not affect diagnosis, monitoring or treatment of the PATIENT, this can be considered not to be cessation or interruption of an intended operation. For example, if in response to the IMMUNITY TEST LEVEL a ventilator stops pumping for 50 ms and then resumes operation such that accuracy is within acceptable limits, this would not be considered cessation or interruption of an intended operation.

Note that it might be necessary to test the ME EQUIPMENT or ME SYSTEM multiple times, e.g. under one set of conditions to assure that it sounds an ALARM signal when it should, within the MANUFACTURER's specifications for sensitivity and response time, and under another set of conditions to assure that it does not sound an ALARM signal when it should not.

Description of the BASIC SAFETY and ESSENTIAL PERFORMANCE including a description how the BASIC SAFETY and ESSENTIAL PERFORMANCE will be monitored against the pass/fail criteria during each test

BASIC SAFETY and ESSENTIAL PERFORMANCE:

This product is a high-tech infra-red (IR) thermometer designed to take human body temperature by measuring the energy of IR emitted from the forehead. The product helps you to assess you and your family members' forehead temperature easily and quickly.

IMMUNITY pass/fail criteria

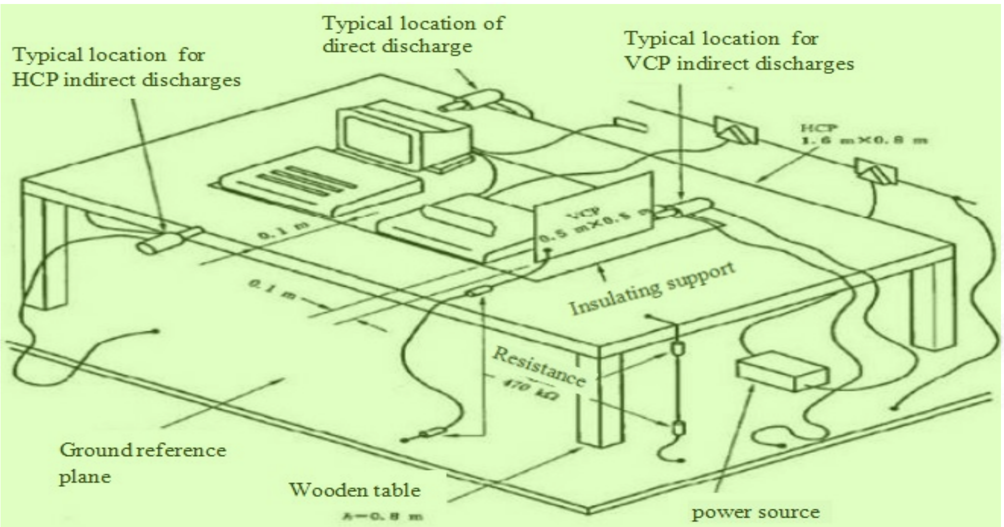
For each immunity test, the EUT should meet following criteria during and after the test.

immunity Pass/Fail Criteria		
Test Description	Pass/Fail Criteria description	Means of monitoring
Electrostatic Discharges	During the immunity tests, the EUT is tested in UE1 mode. The basic safety performance of the tested product: Measurement accuracy: ±0.2℃ at ranges 35.0℃ ~ 42.0℃,other ranges ±0.3℃.The working condition of the EUT is visually inspected by the engineer, which is set at 37℃.The temperature is continuously measured and compared with the blackbody temperature. The accuracy of the display temperature of the EUT does not exceed ±0.2℃.	Visual
Radiated RF EM Fields		
Proximity fields from RF wireless communications		
Note 1: Specific, detailed immunity pass/fail criteria, shall be based on applicable part two standards or risk management, for immunity with regard to em disturbances. These pass/fail criteria shall be included in the risk management file		

6.2 ESD

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	IEC 61000-4-2
Test Date:	2024-07-29
Temperature and Humidity:	28℃ 44% 101kPa
Power Supply:	DC 3V
Discharge Voltage:	Air Discharge: ±2, 4, 8,15kV
	Contact Discharge: ±8 kV
	VCP, HCP: ±8 kV
Polarity:	Positive / Negative
Number of Discharge:	Minimum 10 times at each test point
Discharge Mode:	Single Discharge
Discharge Period:	1 second minimum

6.2.1 Test Setup and Procedure



1. Contact discharge was applied only to conductive surfaces of the EUT. Air discharge was applied only to non-conducted surfaces of the EUT.
2. The EUT was put on a 0.8m high wooden table for table-top equipment or 0.1m high for floor standing equipment standing on the ground reference plane (GRP).
3. A horizontal coupling plane(HCP) 1.6m by 0.8m in size was placed on the table, and the EUT with its cables were isolated from the HCP by an insulating support thick than 0.5mm. The VCP 0.5m by 0.5m in size while HCP were constructed from the same material type and thickness as that of the GRP, and connected to the GRP via a 470kΩ resistor at each end. The distance between EUT and any of the other metallic surfaces except the GRP, HCP and VCP was greater than 1m.
4. During the contact discharges, the tip of the discharge electrode touched the EUT before the discharge switch is operated. During the air discharges, the round discharge tip of the discharge electrode was approached as fast as possible to touch the EUT.
5. After each discharge, the ESD generator was removed from the EUT, the generator is then retriggered for a new single discharge. For ungrounded product, a discharge cable with two resistances was used after each discharge to remove remnant electrostatic voltage. 10 times of each polarity single discharge were applied to HCP and VCP.

6.2.2 Test Results

Test Point:

1. All insulated enclosure / seams, e.g.: Key-press, nonmetallic enclosure of adaptor and main unit,LCD display, seams around enclosure and cables(Refer to the A started arrow in the test setup photo).
2. All accessible metal parts of the enclosure, e.g.: metallic part of the cuff (Refer to the C started arrow in the test setup photo).

Test mode: Mode 1,Mode 2

Air Discharge	Discharge Level (kV)								interval（s）	Test Results
	2		4		8		15			
Test Point	+	-	+	-	+	-	+	-		
A1	ND	ND	ND	ND	ND	ND	ND	ND	1	A
A2	ND	ND	ND	ND	ND	ND	ND	ND	1	A
A3	ND	ND	ND	ND	ND	ND	ND	ND	1	A
A4	ND	ND	ND	ND	ND	ND	ND	ND	1	A
A5	ND	ND	ND	ND	ND	ND	ND	ND	1	A
A6	–	–	–	–	–	–	–	–	–	–
Remark： √：Pass；×：Fail； ND：No Discharge；−：Non-test										

Contact Discharge	Discharge Level (kV)								interval（s）	Test Results
	2		4		6		8			
Test Point	+	-	+	-	+	-	+	-		
C1	–	–	–	–	–	–	–	–	–	–
C2	–	–	–	–	–	–	–	–	–	–
C3	–	–	–	–	–	–	–	–	–	–
Remark： √：Pass；×：Fail； ND：No Discharge；−：Non-test										

Indirect Application	Discharge Level (kV)								interval (s)	Test Results
	2		4		6		8			
Coupling plate - sample direction	+	-	+	-	+	-	+	-		
Horizontally coupled plate - Front	-	-	-	-	-	-	√	√	1	A
Horizontally coupled plate - Left	-	-	-	-	-	-	√	√	1	A
Horizontally coupled plate - Rear	-	-	-	-	-	-	√	√	1	A
Horizontally coupled plate - Right	-	-	-	-	-	-	√	√	1	A
Vertical coupling plate - Front	-	-	-	-	-	-	√	√	1	A
Vertical coupling plate - Left	-	-	-	-	-	-	√	√	1	A
Vertical coupling plate - Rear	-	-	-	-	-	-	√	√	1	A
Vertical coupling plate - Right	-	-	-	-	-	-	√	√	1	A
Remark： √：Pass；×：Fail； ND：No Discharge；－：Non-test										

Remarks:

A: During and after test, The EUT is tested in UE1 mode,the collection point of the temperature sensor is aligned with the blackbody, which is set at 37℃.The temperature is continuously measured and compared with the blackbody temperature. The accuracy of the display temperature of the EUT does not exceed ± 0.2℃.

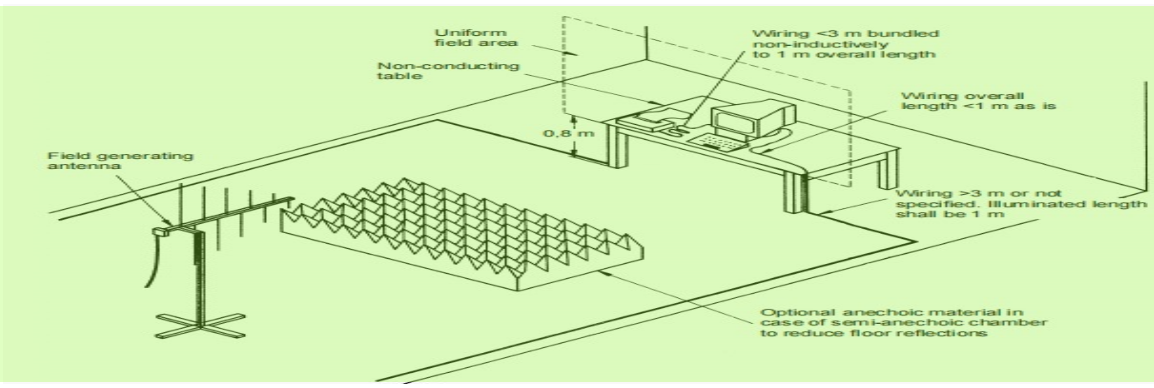
During and after test, the EUT could operate as intended and no disturbance of function. The EUT could meet the requirements of clause 6.1.

N/A: Not applicable (floor mounted EUT or not requested by Standard).

6.3 Radiated Immunity

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	IEC 61000-4-3
Test Date:	2024-07-09/10/24-26
Temperature and Humidity:	25-31℃ 32%-50%
Power Supply:	DC 3V
Frequency Range:	80 MHz to 2.7 GHz
Antenna Polarization:	Horizontal / Vertical
Test level:	10 V/m
Modulation:	80%, 1KHz Amplitude Modulation

6.3.1 Test Setup and Procedure



1. For table-top equipment, the EUT was placed in the chamber on a non-conductive table 0.8m high. For

arrangement of floor-standing equipment, the EUT was mounted on a non-conductive support 0.1m above the supporting plane.

2. If possible, a minimum of 1 m of cable is exposed to the electromagnetic field. Excess length of cables interconnecting units of the EUT shall be bundled low-inductively in the approximate center of the cable to form a bundle 30 cm to 40 cm in length.
3. The EUT was initially placed with one face coincident with the calibration plane. The EUT face being illuminated was contained within the UFA (Uniform Field Area).
4. The frequency ranges to be considered were swept with the signal modulated and pausing to adjust the RF signal level or to switch oscillators and Antennas as necessary. Here the frequency range was swept incrementally, the step size was not exceed 1% of the preceding frequency value.
5. The dwell time of the amplitude modulated carrier at each frequency was not be less than the time necessary for the EUT to be exercised and to respond, and was not less than 0.5 s.
6. The test normally was performed with the generating Antenna facing each side of the EUT.
7. The polarization of the field generated by each Antenna necessitates testing each selected side twice, once with the Antenna positioned vertically and again with the Antenna positioned horizontally.
8. The EUT was performed in a configuration to actual installation conditions, a video camera and/or a audio monitor were used to monitor the performance of the EUT.

6.3.2 Test Results

Frequency	Level	Modulation	Test Mode	Antenna Polarization	EUT Face	Result / Observations
80 MHz to 2.7GHz	10 V/m	1KHz 80% Amp.Mod, 1% increment	Mode 1,2.	V	Front	A
				H		A
				V	Rear	A
				H		A
				V	Left	A
				H		A
				V	Right	A
				H		A
				N/A	Top*	N/A
				N/A		N/A
				N/A	Bottom*	N/A
				N/A		N/A

Remarks:

Front: the front of the EUT faces to transmitting Antenna (refer to Radiated Immunity test setup photo)

A: During and after test,The EUT is tested in UE1 mode,the collection point of the temperature sensor is aligned with the blackbody, which is set at 37℃.The temperature is continuously measured and compared with the blackbody temperature. The accuracy of the display temperature of the EUT does not exceed ± 0.2℃.

During and after test, the EUT could operate as intended and no disturbance of function. The EUT could meet the requirements of clause 6.1.

*: Tests against top side and bottom side are only for adaptor.

The EUT does meet the Radiated Immunity requirements of the Standard.

6.4 Proximity Fields from RF Wireless Communications Equipment

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	IEC 61000-4-3
Test Date:	2024-07-09/10/24-26
Temperature and Humidity:	25-31℃ 32%-50%
Power Supply:	DC 3V
Test Level:	Table 9 of IEC 60601-1-2
Antenna Polarization:	Horizontal / Vertical
Modulation:	Pulse Modulation,18Hz or 217Hz

6.4.1 Test Setup and Procedure

1. For table-top equipment, the EUT was placed in the chamber on a non-conductive table 0.8m high. For arrangement of floor-standing equipment, the EUT was mounted on a non-conductive support 0.1m above the supporting plane.
2. If possible, a minimum of 1 m of cable is exposed to the electromagnetic field. Excess length of cables interconnecting units of the EUT shall be bundled low-inductively in the approximate center of the cable to form a bundle 30 cm to 40 cm in length.
3. The EUT was initially placed with one face coincident with the calibration plane. The EUT face being illuminated was contained within the UFA (Uniform Field Area).
4. The test distances between Antenna and EUT are 3m for the frequencies between 1970MHz and 5785MHz and 2.5m for the frenquencies between 385MHz and 1970MHz .
5. The dwell time of the amplitude modulated carrier at each frequency was not be less than the time necessary for the EUT to be exercised and to respond, and was not less than 1s.
6. The test normally was performed with the generating Antenna facing each side of the EUT.
7. The polarization of the field generated by each Antenna necessitates testing each selected side twice, once with the Antenna positioned vertically and again with the Antenna positioned horizontally.
8. The EUT was performed in a configuration to actual installation conditions, a video camera and/or a audio monitor were used to monitor the performance of the EUT.

6.4.2 Test result

Frequency (MHz)	Band (MHz)	Level	Modulation	Test mode	Antenna Polarization	EUT face	Result/ observations
385	380-390	27	Pulse modulation 18Hz	Mode 1,2.	Horizontal / Vertical	Front, Rear, Left, Right,	A
450	430-470	28					
710	704-787	9	Pulse modulation 217Hz				
745							
780							
810	800-960	28	Pulse modulation 18Hz				
870							
930							
1720	1700-1990	28	Pulse modulation 217Hz				
1845							
1970							
2450	2400-2570	28					
5240	5100-5800	9					
5500							
5785							

Remarks:

A: During and after test, The EUT is tested in UE1 mode,the collection point of the temperature sensor is aligned with the blackbody, which is set at 37 ℃.The temperature is continuously measured and compared with the blackbody temperature. The accuracy of the display temperature of the EUT does not exceed ±0.2℃.

During and after test, the EUT could operate as intended and no disturbance of function. The EUT could meet the requirements of clause 6.1.

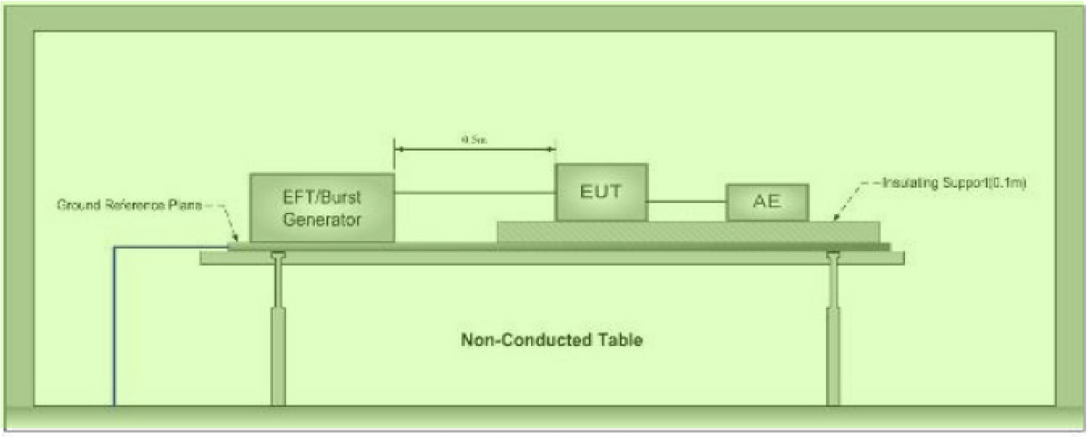
*: Tests against top side and bottom side are only for adaptor.

The EUT does meet the immunity requirements of Proximity Fields from RF Wireless Communications Equipment.

6.5 Electrical Fast Transients (EFT)

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	N/A
Test Date:	N/A
Temperature and Humidity:	N/A
Power Supply:	N/A
Test Level:	N/A
Polarity:	N/A
Repetition Frequency:	N/A
Burst Duration:	N/A
Test Duration:	N/A

6.5.1 Test Setup and Procedure



Ground Reference Plane

1. The EUT was placed on a ground reference plane(GRP) insulated by an insulating support 0.1m thick and the GRP was placed on a 0.8m high wooden table for table-top equipment. For floor standing equipment, the EUT was placed on a 0.1m high wooden support above the GRP.
2. The GRP shall project beyond the EUT and the clamp by at least 0.1m on all sides. The distance between the EUT and any other of the metallic surface except the GRP was greater than 0.5m. All cables to the EUT was placed on the insulation support 0.1m above GRP. Cables not subject to EFT was routed as far as possible from cable under test to minimize the coupling between the cables.
3. The length of signal and power cable between the EUT and EFT generator was 0.5m. If the cable is a non-detachable supply cable more than 0.5m, the excess length of this cable shall be folded to avoid a flat coil and situated at a distance of 0.1m above the GRP.
4. The EUT was conducted the below specified level voltage test for line to neutral or line to neutral to earth(for clamp coupling is for the signal line), 120 seconds duration. If the equipment contains identical ports, only one was tested; multiconductor cables, such as a 50-pair telecommunication cable, was tested as a single cable. Cables did not be split or divided into groups of conductors for this test; interface ports, which were intended by the manufacturer to be connected to data cables not longer than 3 m, did not be tested.
5. During the test, the EUT was tested with the hand-held parts terminated with artificial hand.

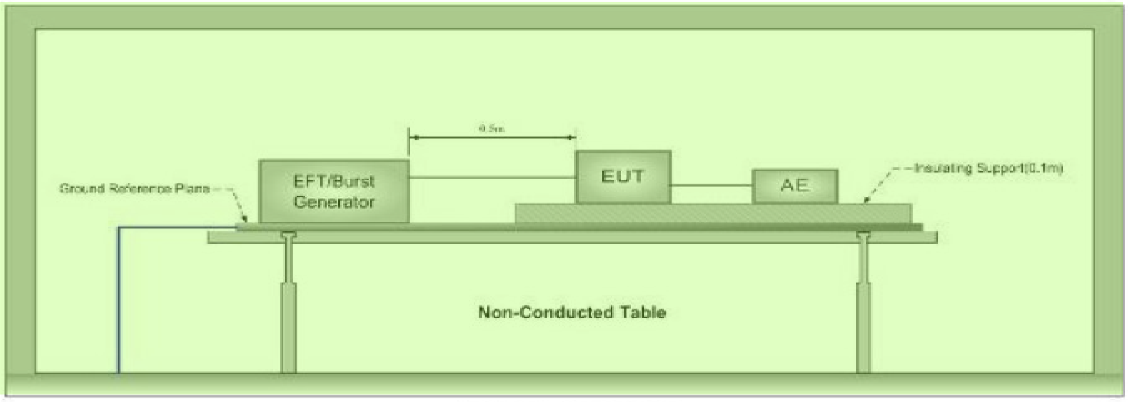
6.5.2 Test Results

Note: The EUT is powered by batteries , and does not apply to this clause.

6.6 Surge

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	N/A
Test Date:	N/A
Temperature and Humidity:	N/A
Power Supply:	N/A
Test Level:	N/A
Polarity:	N/A
Generator source impedance:	N/A
Trigger Mode:	N/A
No. of surges:	N/A

6.6.1 Test Setup and Procedure



Ground Reference Plane

1. The EUT was placed on a ground reference plane(GRP) insulated by an insulating support 0.1m thick and the GRP was placed on a 0.8m high wooden table for table-top equipment. For floor standing equipment, the EUT was placed on a 0.1m high wooden support above the GRP.
2. The 1.2/50 μs surge was to be applied to the EUT power supply Terminals via the capacitive coupling network .Decoupling networks were required in order to avoid possible adverse effects on equipment not under test that may be powered by the same lines and to provide sufficient decoupling impedance to the surge wave so that the specified wave may be applied on the lines under test.
3. The power cord between the EUT and the coupling/decoupling network do not exceed 2m in length. The interconnection line between the EUT and the coupling/ decoupling network shall not exceed 2m in length.
4. The EUT was conducted 1kV Test voltage: for line to line and line to neutral and conducted 2kV Test voltage: for line to earth and neutral to earth, five positive pulses and five negative pulses each at 90° and 270° for AC power ports and five positive pulses and five negative surge pulses for DC power ports. The test levels were applied on the EUT with a 2Ω generator source impedance for power supply Terminals and 40Ω output impedance for interconnection lines. The tests were done at repetition rate 1 per minute.

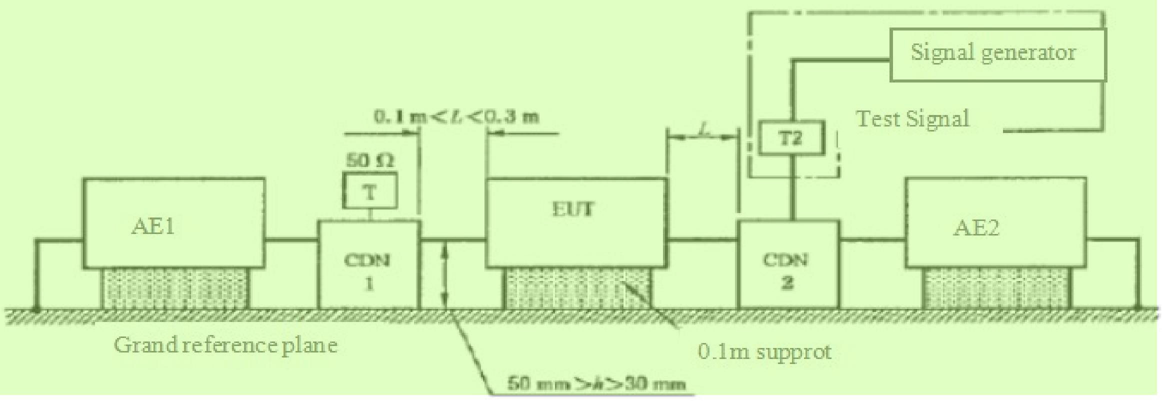
6.6.2 Test Results

Note: The EUT is powered by batteries , and does not apply to this clause.

6.7 Conducted Immunity 0.15 MHz to 80 MHz

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	N/A
Test Date:	N/A
Temperature and Humidity:	N/A
Power Supply:	N/A
Frequency Range:	N/A
Test level:	N/A
Modulation:	N/A
Dwell Time:	N/A

6.7.1 Test Setup and Procedure



1. The EUT was placed on an insulating support of 0.1m height above a ground reference Plane,arranged and connected to satisfy its functional requirement. All cables exiting the EUT was supported at a height of at least 30 mm above the ground reference plane.
2. The coupling and decoupling devices were required, they were located between 0.1m and 0.3m from the EUT. This distance was to be measured horizontally from the projection of the EUT on to the ground reference plane to the coupling and decoupling device.
3. Each AE, used with clamp injection, shall be placed on an insulating support 0.1m above the ground reference plane. A decoupling network shall be installed on each cable between the EUT and AE except the cable under test. All cables connected to each AE, other than those being connected to the EUT, shall be provided with decoupling networks. The decoupling networks connected to each AE (except those on cables between the EUT and AE) shall be applied no further than 0.3m from the AE. The cable(s) between the AE and the decoupling network (s) or in between the AE and the injection clamp shall not be bundled nor wrapped and shall be kept between 30 mm and 50 mm above the ground reference plane.
4. The frequency range was swept from 150 kHz to 80 MHz, using the signal levels established during the setting process, and with the disturbance signal 80% amplitude modulated with a 1kHz sine wave, pausing to adjust the RF signal level or to change coupling devices as necessary. Where the frequency was swept incrementally, the step size do not exceed 1% of the preceding frequency value. The dwell time of the amplitude modulated carrier at each frequency was not less than the time necessary for the EUT to be exercised and to respond, and was not less than 0.5 s.
5. During the test, the EUT was tested with the hand-held parts terminated with artificial hand.

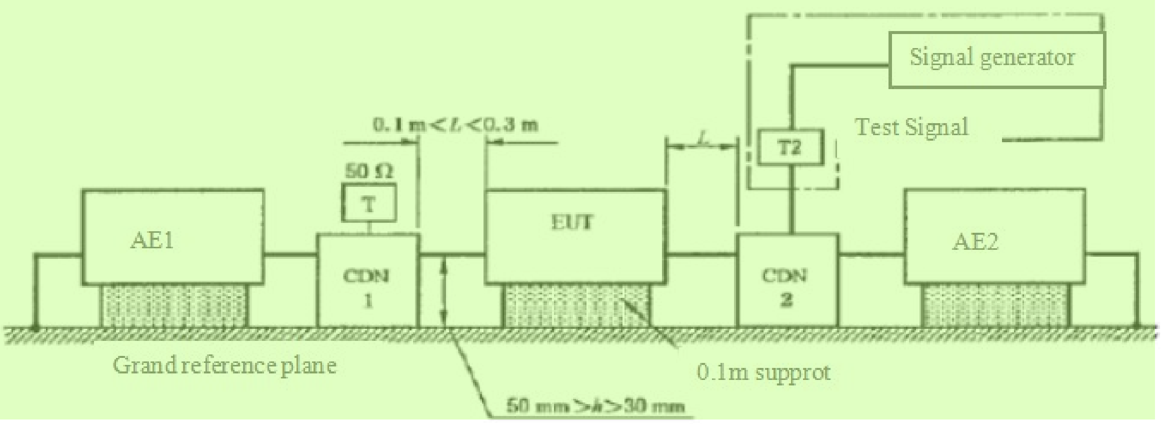
6.7.2 Test Results

Note: The EUT is powered by batteries , and does not apply to this clause.

6.8 Voltage Dips and Interruptions

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	N/A
Test Date:	N/A
Temperature and Humidity:	N/A
Power Supply:	N/A
Test Level:	N/A
No. of Dips / Interruptions:	N/A

6.8.1 Test Setup and Procedure



1. The EUT was placed on a ground reference plane(GRP) insulated by an insulating support 0.1m thick and the GRP was placed on a 0.8m high wooden table for table-top equipment. For floor standing equipment, the EUT was placed on a 0.1m high wooden support above the GRP.
2. The test was performed with the EUT connected to the test generator with the shortest power supply cable as specified by the EUT manufacturer.
3. The EUT was tested for each selected combination of test level and duration with a sequence of three dips/interruptions with intervals of 10 s minimum. Each representative mode of operation was tested.
4. For EUT with more than one power cord, each power cord was tested individually.

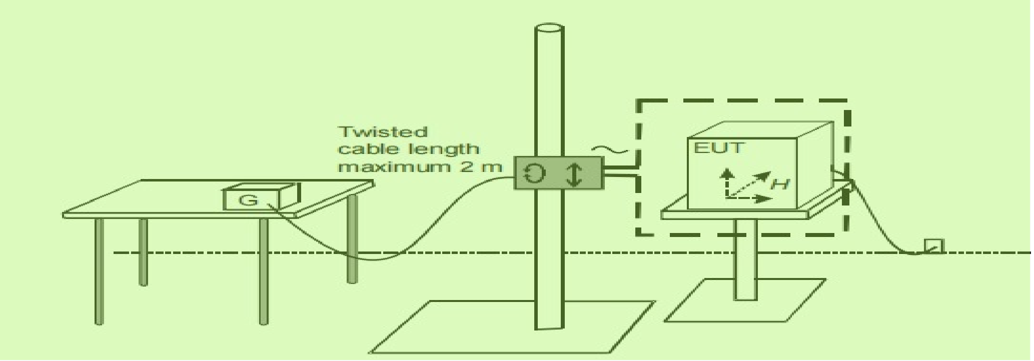
6.8.2 Test Results

Note: The EUT is powered by batteries , and does not apply to this clause.

6.9 Power Frequency Magnetic Field Immunity

Test Requirement:	IEC 60601-1-2:2014+A1:2020
Test Method:	N/A
Test Date:	N/A
Temperature and Humidity:	N/A
Power Supply:	N/A
Field frequency:	N/A
Face under Test:	N/A
Test level:	N/A

6.9.1 Test Setup and Procedure



1. The EUT was placed on a ground reference plane(GRP) insulated by an insulating support 0.1m thick and the GRP was placed on a 0.8m high wooden table for table-top equipment. For floor standing equipment, the EUT was placed on a 0.1m high wooden support above the GRP.
2. The EUT was configured and connected to satisfy its functional requirements. The power supply, input and output circuits was connected to the sources of power supply, control and signal. The cables supplied or recommended by the equipment manufacturer were used. In absence of any recommendation, unshielded cables were adopted, of a type appropriate for the signals involved. All cables were exposed to the magnetic field for 1m of their length.
3. The EUT was subjected to the test magnetic field by using the induction coil of standard dimensions (1m×1m) by the immersion method.
4. The induction coil was then be rotated by 90^0 in order to expose the EUT to the test field with different orientations. For floor-standing equipment the test was repeated by moving and shifting the induction coils, in order to test the whole volume of the EUT for each orthogonal direction.

6.9.2 Test Results

IEC 60601-1-2:2014+A1:2020/EN 60601-1-2:2015+A1:2021 Table 4 Remark d Applies only to ME EQUIPMENT and ME SYSTEMS with magnetically sensitive components or circuitry.
The EUT does not contain magnetic sensing components or circuits (see Annex statement) and exempt from this test

7 Identifiers, tags, and files

IEC 60601-1-2:2014+A1:2020			
Clause	Requirement + Test	Result - Remark	Verdict
4	GENERAL REQUIREMENTS		
4.1	risks resulting from reasonably foreseeable electromagnetic disturbances taken into account in the risk management process.	RMF Reference Document: PT9L-FXPJ01 V1.0	P
4.2	Non-me equipment used in an me system		N/A
	Check 16.1 of general standard, checked by inspection of therisk management file and objective evidence of compliance with the respective emc standards, or by the tests of this collateral standard	EUT does not contain non-ME equipment.	N/A
	non-me equipment used in an me system complies with iec andiso emc standards applicable to that equipment, checked by inspection of the risk management file and objective evidenceof compliance with the respective emc standards, or by the tests of this collateral standard	EUT does not contain non-ME equipment.	N/A

Clause	Requirement + Test	Result - Remark	Verdict
	non- me equipment used in an me system for which the intendedem environment could result in the loss of basic safety oressential performance of the me system due to the non-me equipment tested according to the requirements of this collateral standard, checked by inspection of the RISKmanagement file and objective evidence of compliance with the respective EMC standards, or by the tests of this collateral standard	EUT does not contain non-ME equipment.	N/A
4.3.1	Configurations		P
	me equipment and me systems tested in representative configurations, consistent with intended use, that are most likely to result in unacceptable risk as determined by themanufacturer (This was determined using risk analysis, experience, engineering analysis, or pretesting). Compliance checked by inspection of the test report and the risk management file.	See Chapter 5.3 and RMF Reference Document: PT9L-FXPJ01 V1.0	P
4.3.3	Power input and frequencies	See Chapter 5.2 for reference	P
5	IDENTIFICATION, MARKING AND DOCUMENTS		
5.1	Additional requirements for marking on the outside of me equipment and me systemsspecified for use only in a shielded location special environment		N/A
	me equipment and me systems specified for use only in a shielded location special environment labelled with a clearly legible warning that they should be used only in the specified type of shielded location	The EUT does not specified for use only in a shielded location special environment	N/A
5.2	ACCOMPANYING DOCUMENTS		
5.2.1	Instructions for use		
5.2.1.1	General		
a)	A statement of the environments for which the me equipment orme system is suitable. Relevant exclusions determined byrisk analysis, are listed.	Reference Document: PT9L-SMSY01 V1.0	P
b)	The essential performance of me equipment and a description of what the operator can expect if the essential performanceis lost or degraded due to EM disturbances.	Reference Document: PT9L-SMSY01 V1.0	P
c)	A warning regarding stacking and location close to otherequipment	Reference Document: PT9L-SMSY01 V1.0	P

IEC 60601-1-2:2014+A1:2020			
Clause	Requirement + Test	Result - Remark	Verdict
d)	List of cables, transducers and accessories	Reference Document: PT9L-SMSY01 V1.0	P
e)	A warning that other cables and accessories may negatively affect EMC performance	Reference Document: PT9L-SMSY01 V1.0	P
f)	A statement that portable RF communications equipmentincluding Antennas, can effect medical electrical equipment. The warning includes a use distance such as "…be used no closer than 30 cm (12 inches) to any part of the [me equipmentor me system], including cables specified by manufacturer"	Reference Document: PT9L-SMSY01 V1.0	P
5.2.1.2	Requirements applicable to me equipment and me systems classified class A according to CISPR 11		N/A
	For me equipment and me systems that are classified as class A according to CISPR 11, the instructions for use include the following note: note: "The emissions characteristics of this equipment make it suitable for use in industrial areas and hospitals (CISPR 11 class A). If it is used in a residential environment (for which CISPR 11 class B is normally required) this equipment might not offer adequate protection to radio-frequency communication services. The user might need to take mitigation measures, such as relocating or re-orienting the equipment."	The EUT complaints class B limit	N/A
5.2.2	Technical description		
5.2.2.1	Requirements applicable to all me equipment and me systems		
	The technical description describes precautions to be taken to prevent adverse events to the patient and Operator due to electromagnetic disturbances	Reference Document: Instructions for Use PT9L-SMSY01 V1.0	P
a)	Compliance for each emissions and immunity standard or test specified by this collateral standard, e.g. emissions class and group and immunity test level	Reference Document: Instructions for Use PT9L-SMSY01 V1.0	P

IEC 60601-1-2:2014+A1:2020			
Clause	Requirement + Test	Result - Remark	Verdict
b)	Any deviations from this collateral standard and allowances used	Reference Document: N/A. Full tests were performed on the EUT. No deviation from this collateral standard and allowances used.	N/A
c)	All necessary instructions for maintaining basic safety andessential performance with regard to electromagnetic disturbances for the expected service life	Reference Document: Instructions for Use PT9L-SMSY01 V1.0	P
5.2.2.2	Requirements applicable to me equipment specified for use only in shielded location Special Environment		
	The technical description includes the following information:		
a)	A warning to the effect that: WARNING: Failure to use this equipment in the specified type of shielded location could result in degradation of performance, interference with other equipment or interference with radio services	The EUT does not intend to use only in shielded location Special Environment.	N/A
b)	Specifications for shielded location including: – minimum RF shielding effectiveness; – for each cable that enters or exits the shielded location, the minimum RF filter attenuation; and – the frequency range(s) over which the specifications apply	Reference Document: N/A. The EUT does not intend to use only in shielded location Special Environment.	N/A
c)	Test methods for measurement of RF shielding effectiveness and RF filter attenuation	The EUT does not intend to use only in shielded location Special Environment.	N/A
d)	One or more of the following and a recommendation that a notice containing this information be posted at the entrance(s) to the shielded location: – a specification of the emissions characteristics of otherequipment allowed inside the shielded location with the me equipment or me system; – a list of specific equipment allowed; – a list of types of equipment prohibited.	Reference Document: N/A. The EUT does not intend to use only in shielded location Special Environment.	N/A

IEC 60601-1-2:2014+A1:2020			
Clause	Requirement + Test	Result - Remark	Verdict
5.2.2.3	Requirements applicable to me equipment that intentionally receive RF electromagnetic energy include the following information... : - each frequency or frequency of reception, - the preferred frequency or frequency band, if applicable, and - the bandwidth of the receiving section of the ME Equipment in those bands	Reference Document: N/A Equipment does not intentionally receive RF electromagnetic energy	N/A
5.2.2.4	Requirements applicable to the me equipment that include RF transmitters the technical description includes the frequency or frequency band of transmission, the type and frequency characteristics of the modulation and the effective radiated power(ERP).. :	EUT does not include RF emission	N/A
5.2.2.5	Requirements applicable to Permanently Installed Large me equipment and large me systems		
	The technical description includes the following information:		
a)	A statement that an exemption has been used and that theequipment has not been tested for radiated rf immunity over the entire frequency range 80 MHz to 6 GHz	The EUT is not a permanently installed large ME systems.	N/A
b)	WARNING: “This equipment has been tested for radiated rf immunity only at selected frequencies, and use nearby of emitters at other frequencies could result in improper operation”	The EUT is not a permanently installed large ME systems.	N/A
c)	A list of the frequencies and modulations used to test theimmunity of the me equipment or me systems	The EUT is not a permanently installed large ME systems.	N/A
5.2.2.6	Requirements applicable to me equipment that claim compatibility with HF Surgicalequipment		N/A
	Technical description includes a statement of HF Surgical Equipment compatibility and the conditions of intended useduring HF Surgery	Reference Document: N/A The EUT does not claim compatibility with HF Surgical equipment	N/A

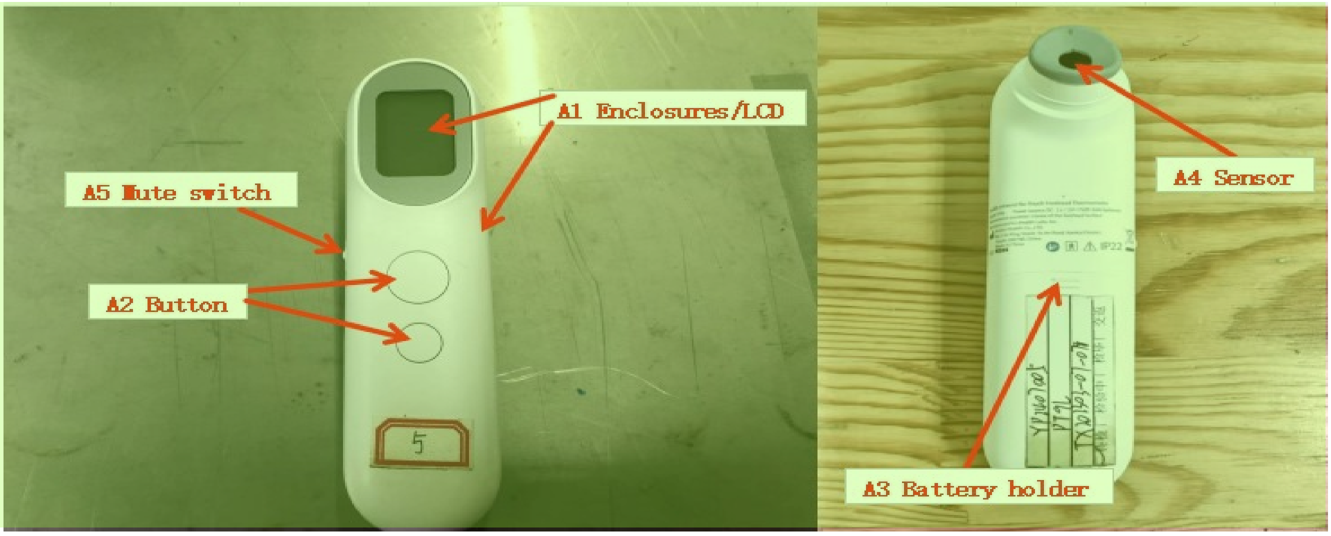
8 Photographs

8.1 Radiated Emission Test Setup

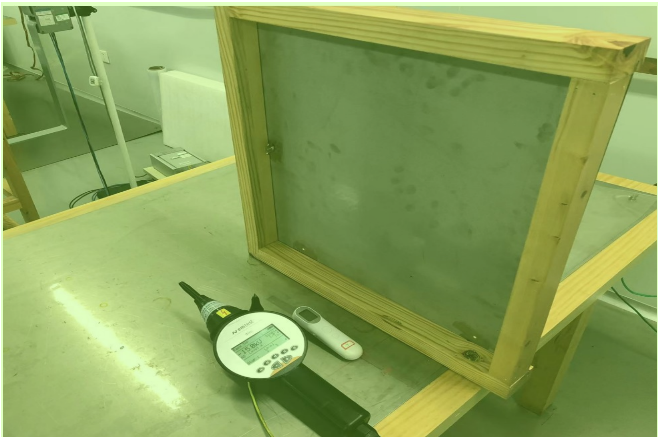
Mode 1

Mode 2

8.2 ESD Test Setup



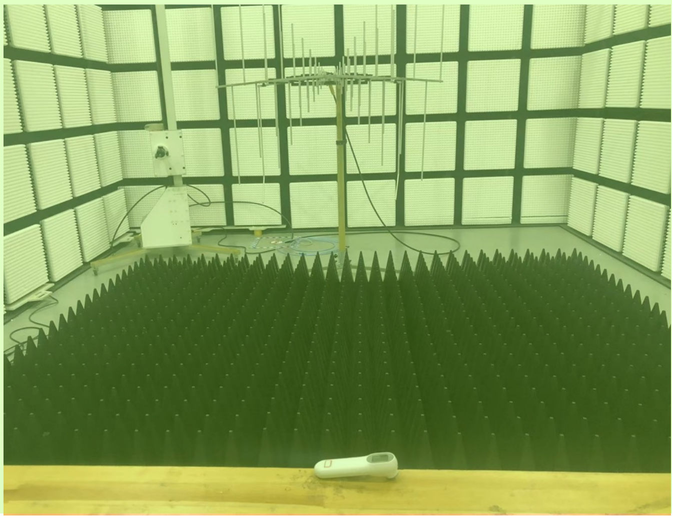
Discharge position



Mode1

Mode 2

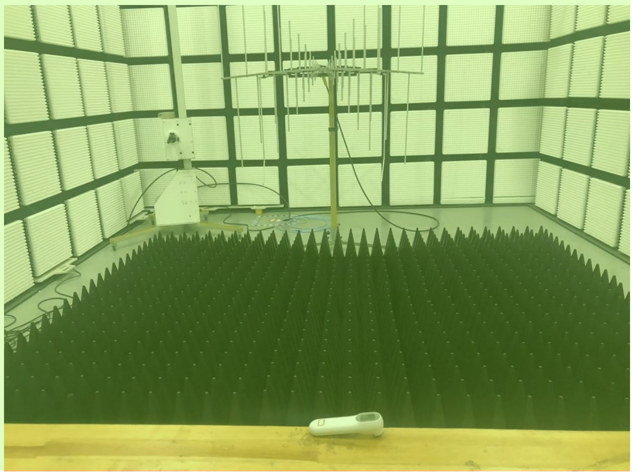
8.3 Radiated Immunity Test Setup



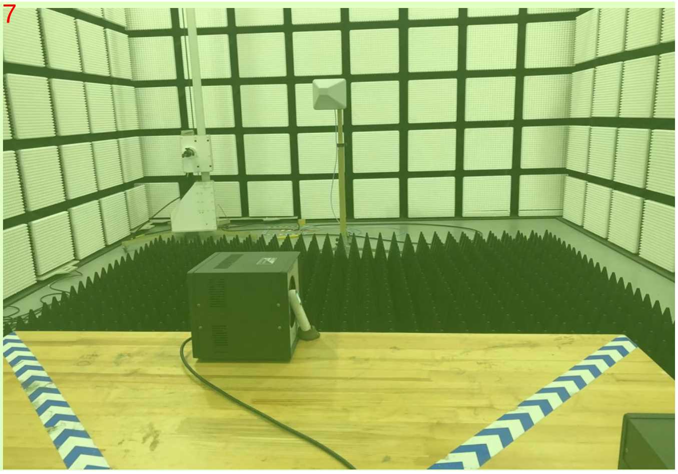
Mode 1

Mode 2

8.4 Proximity Fields from RF Wireless Communications Equipment Test Setup

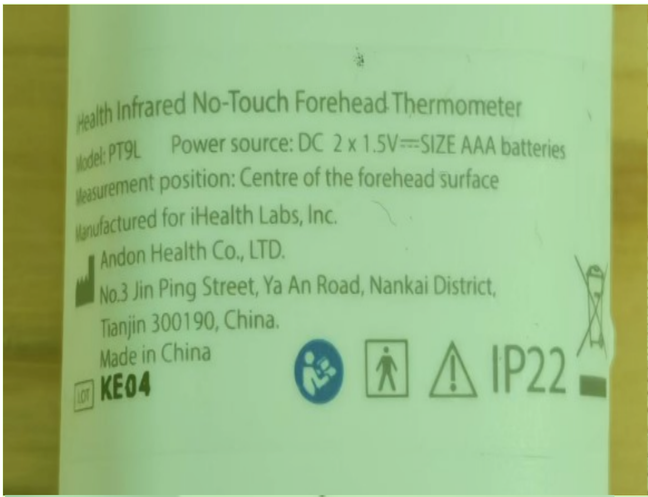


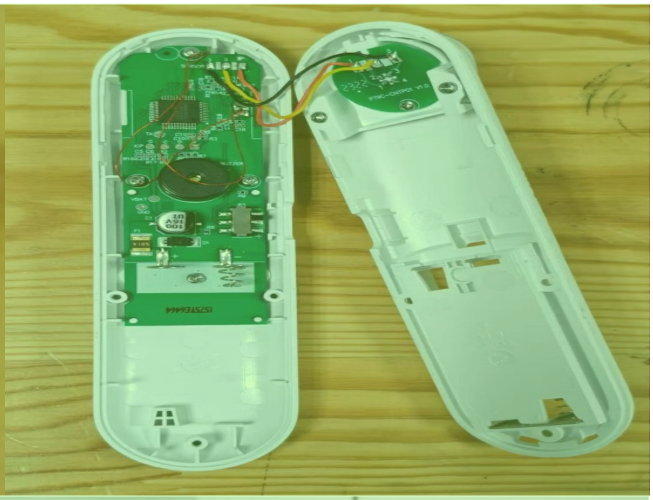
Battery Power 385MHz~1970MHz



Battery Power 2450MHz~5785MHz

8.5 EUT





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